```python
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# Use TkAgg backend for interactive plotting
matplotlib.use('TkAgg')

# ========================
# Load data
# ========================

# restaurants = pd.read_csv('data.csv')  # Example alt dataset (unused)
restaurants = pd.read_csv('restaurants.csv')
reviews = pd.read_csv('reviews.csv')

# ========================
# Clean and enrich review data
# ========================

# Convert review dates to datetime and extract year
reviews['date'] = pd.to_datetime(reviews['date'])
reviews['year'] = reviews['date'].dt.year

# Total count of rows in `reviews['stars']`, repeated in every row
reviews['count'] = reviews['stars'].count()

# Merge reviews with restaurant metadata
reviews_full = reviews.merge(
   restaurants,
   on='business_id',
   how='left'
)

# ========================
# Aggregate by state and year
# ========================

state_year = (
   reviews_full
   .groupby(['state', 'year'])
```

```
    .agg(
        avg_stars=('stars', 'mean'),
        count=('stars', 'size')
    )
    .reset_index()
)

# ==========================
# Filter out sandwich chains of interest
# ==========================

# Subway reviews
subway = reviews_full[
    reviews_full['name'].str.contains('Subway', case=False, na=False)
]
subway.shape
subway['stars'].describe()
# print(subway.shape)
# print(subway['stars'].describe())

# Bonus section: local vs global Subway
subway_local_mean = 3.2
print("subway_local_mean:", subway_local_mean)

subway_global = subway['stars']
subway_global_mean = round(subway_global.mean(), 3)
print("subway_global_mean:", subway_global_mean)

# Jersey Mike's reviews
jersey_mikes = reviews_full[
    reviews_full['name'].str.contains("Jersey Mike's", case=False, na=False)
]
jersey_mikes.shape
jersey_mikes['stars'].describe()
# print(jersey_mikes.shape)
# print(jersey_mikes['stars'].describe())

# Wawa reviews
wawa = reviews_full[
    reviews_full['name'].str.contains('Wawa', case=False, na=False)
]
wawa.shape
```

```
wawa['stars'].describe()
# print(wawa.shape)
# print(wawa['stars'].describe())

# Combine all three sandwich brands
sandwich_places = pd.concat([subway, jersey_mikes, wawa])

# =========================
# Classify restaurant chain types
# =========================

# Count how many distinct cities each brand appears in
city_counts = (
   restaurants
   .groupby('name')['city']
   .nunique()
   .sort_values(ascending=False)
)
# print(city_counts)

# Map each brand to a chain type based on city count
type_map = city_counts.copy()
type_map = np.where(
   city_counts >= 50, 'national',
   np.where(city_counts == 1, 'local', 'other')
)

# Attach chain type to restaurants
restaurants['chain_type'] = restaurants['name'].map(
   dict(zip(city_counts.index, type_map))
)

# Merge chain_type back into full reviews
reviews_full = reviews_full.merge(
   restaurants[['business_id', 'chain_type']],
   on='business_id',
   how='left'
)

# ========================
# Restrict to reviews from 2018 to 2021
# ========================
```

```
reviews_full['year'] = pd.to_datetime(reviews_full['date']).dt.year
reviews_18_21 = reviews_full[reviews_full['year'].between(2018, 2021)]

# ========================
# Plots
# ========================

# 1. Average rating by year
plt.figure('Question 1A')
sns.lineplot(x='year', y='avg_stars', data=state_year)
plt.xlabel('Year')
plt.ylabel('Average star rating')
plt.title('Average Yelp rating by year')
plt.ylim(0, 5)

# 2. Number of reviews by state and year
plt.figure('Question 1B')
sns.barplot(x='state', y='count', data=state_year)
plt.title('Number of Yelp reviews by state and year')

# 3. Boxplot of star ratings for Subway vs competitors
plt.figure('Question 2')
sns.boxplot(x='name', y='stars', data=sandwich_places)
plt.title('Subway Vs Competitors')

# 4. Boxplot by chain size (local vs national vs other)
plt.figure('Question 3')
sns.boxplot(x='chain_type', y='stars', data=reviews_full)
plt.title('Rating by Size')

# 5. Distribution of ratings from 2018 to 2021
plt.figure('Question 4')
sns.countplot(
   x='stars',
   data=reviews_18_21,
   order=sorted(reviews_18_21['stars'].unique())
)
plt.xlabel('Star Rating')
plt.ylabel('Number of Reviews')
plt.title('Ratings from 2018 to 2021')
plt.tight_layout()
```

```
# ========================
# Bonus visualization 1
# Histogram of all Subway ratings
# with local and global means highlighted
# ========================

plt.figure('Bonus Visualization #1', figsize=(6, 4))

plt.hist(
   subway['stars'],
   bins=10,
   edgecolor='black',
   alpha=0.7
)

plt.axvline(
   subway_local_mean,
   color='red',
   linewidth=2,
   label='Local Subway (3.2)'
)

plt.axvline(
   subway_global_mean,
   color='green',
   linewidth=2,
   label=f'Global Mean ({subway_global_mean})'
)

plt.xlabel('Star Rating')
plt.ylabel('Frequency')
plt.title('Distribution of Global Subway Ratings\nwith Local Store Highlighted')
plt.legend()
plt.tight_layout()

# ========================
# Bonus visualization 2
# Subway vs competitors histogram
# ========================

competitors = pd.concat([jersey_mikes, wawa], ignore_index=True)
```

```
plt.figure('Bonus Visualization #2', figsize=(8, 5))

# Subway histogram
plt.hist(
   subway['stars'],
   bins=[1, 2, 3, 4, 5, 6],
   alpha=0.6,
   label='Subway',
   edgecolor='black'
)

# Competitors histogram
plt.hist(
   competitors['stars'],
   bins=[1, 2, 3, 4, 5, 6],
   alpha=0.55,
   label="Jersey Mike's + Wawa",
   edgecolor='black'
)

plt.xlabel('Star Rating')
plt.ylabel('Number of Reviews')
plt.title('Distribution of Ratings: Subway vs Competitors')
plt.xticks([1, 2, 3, 4, 5])
plt.legend()
plt.tight_layout()

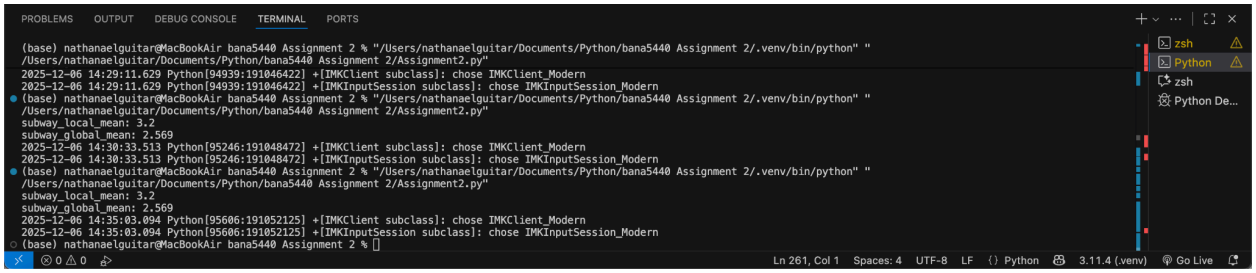
# =========================
# Bonus visualization 3
# Distribution of average rating per Subway store
# =========================

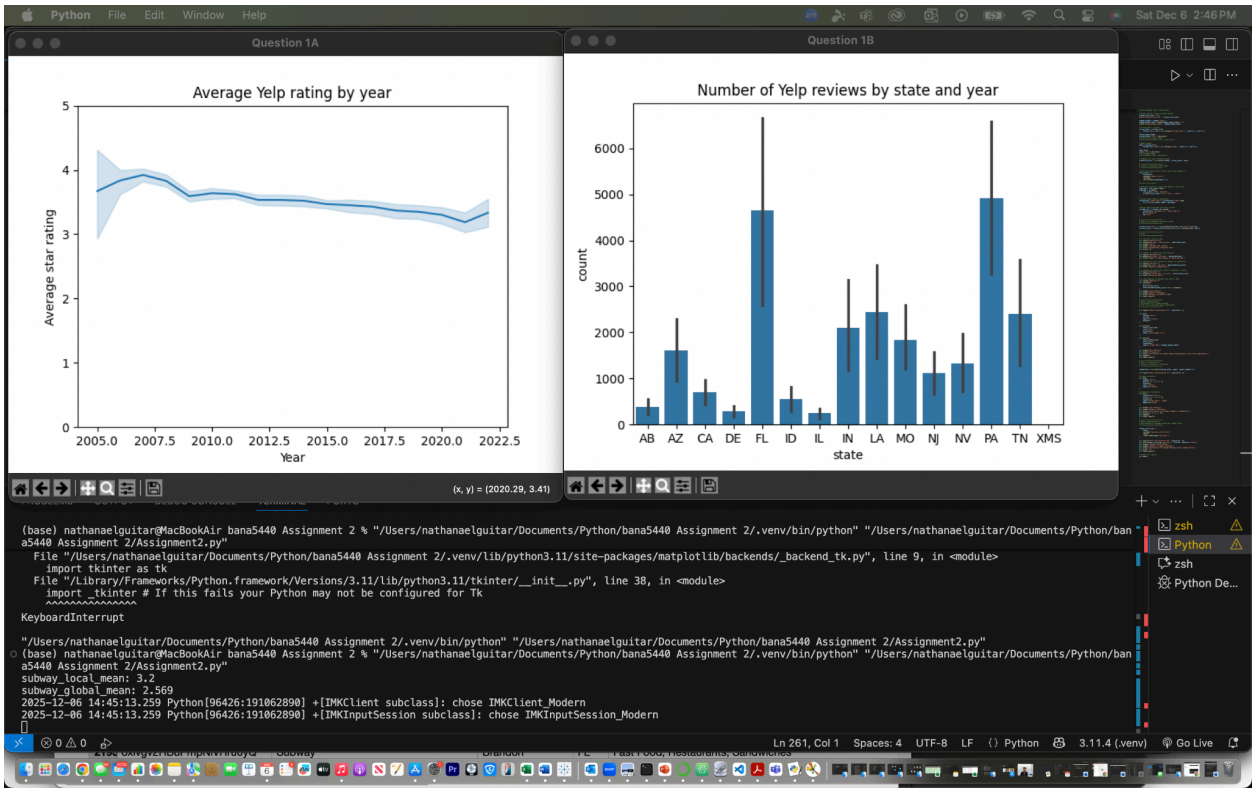
subway_store_avg = (
   subway
   .groupby('business_id')['stars']
   .mean()
   .reset_index(name='avg_stars')
)

plt.figure('Bonus Visualization #3', figsize=(6, 4))
plt.hist(subway_store_avg['avg_stars'], bins=10, edgecolor='black')
```

```
plt.xlabel('Average Store Rating')
plt.ylabel('Number of Subway Locations')
plt.title('Distribution of Average Ratings Across Subway Stores')
plt.xlim(1, 5)
plt.tight_layout()

# Render all figures
plt.show()
```






Python File Edit Window Help
Sat Dec 6 2:46 PM
Question 3
Rating by Size
stars
chain_type
local
national
Question 4
Ratings from 2018 to 2021
Number of Reviews
Star Rating
Screenshot 025-0...58PM (2)
Screenshot 2025-07...0PM (2)
Screenshot 2026-07...8PM (2)
Screenshot 2025-0....43.03PM
sqlite-tools-osx-x64-3490200
Bonus Visualization #1
Distribution of Global Subway Ratings with Local Store Highlighted
Local Subway (3.2)
Global Mean (2.569)
Frequency
Star Rating
Question 2
Subway Vs Competitors
stars
Subway
Jersey Mike's Subs
Wawa
name
Distribution of Ratings: Subway vs Competitors
Subway
Jersey Mike's + Wawa
Number of Reviews
Star Rating
Bonus Visualization #3
Distribution of Average Ratings Across Subway Stores
Number of Subway Locations
Average Store Rating